

Haoyu Zhang

📍 Chongqing, China ✉ zhanghaoyu@cqcai.edu.cn ☎ 152 2139 9427 🌐 haoyuuzhang.github.io

Education

- | | |
|---|-----------------------------------|
| MS Southwest University , Operations Research
• Top 5%(3/65): Optimization, Variational Analysis, etc. | Chongqing
Sept 2025 – present |
| BS Donghua University , Mathematics and Applied Mathematics
• Top 14%(8/61): Mathematical Modeling, Discrete Mathematics, Data Structures and Algorithms, etc. | Shanghai
Sept 2021 – June 2025 |

Work Experience

- | | |
|---|---|
| Appen Ltd. , Math AIGC Training Intern
• Designed Chain-of-Thought (CoT) structures to improve the logical consistency and interpretability of Large Language Models (LLMs) in solving complex mathematical problems.
• Participated in establishing the company's first set of data annotation specifications for mathematical reasoning chains, achieving structured reasoning paths and unified annotation standards. | Chongqing, China
Jan 2025 – Apr 2025 |
|---|---|

Research Experience

- | | |
|--|----------------------|
| Fluid Dynamics and Thermal Coupling Modeling in Greenhouse Environments
Group Leader
• Established 3D wind speed and temperature distribution models for glass greenhouses based on Navier-Stokes (N-S) and heat conduction equations, accounting for heat loss from plant transpiration and correcting for complex airflow in crop scenarios.
• Developed a numerical solver using finite difference and control volume methods to discretize and solve complex PDEs and visualize distribution results, providing a modeling basis for intelligent agricultural environment control. | Nov 2023 – Apr 2024 |
| Mathematical Modeling and Research of Mixed Human-Machine Traffic Flow
Independent Researcher (Advisor: Prof. Linglong Du)
• Developed a mathematical model for mixed traffic flow of human-driven and autonomous vehicles in ring-road scenarios, incorporating the Bando-Follow the Leader (Bando-FTL) model and accounting for stochastic slowdown effects.
• Introduced a Proportional-Integral (PI) controller for Autonomous Vehicles (AVs) to optimize regulation capabilities. Matlab simulations confirmed the system effectively increased average speed and reduced variance, thereby alleviating congestion and reducing energy consumption. | Sept 2024 – Apr 2025 |

Projects and Coursework

- | | |
|---|----------------------|
| Climbing Posture Analysis and Coordination Optimization Based on Computer Vision
Course Project
• Utilized OpenPose for precise recognition and tracking of human key points across three types of climbing scenarios.
• Designed a frame counter to calculate displacement via discrete accumulation, visualized motion curves, and computed a synchronization index through phase domain transformation to evaluate athlete coordination. | Sept 2023 – Jan 2024 |
|---|----------------------|

Nighttime Vehicle Speed Estimation Model Based on Optical Flow Analysis

Mar 2024 – June 2024

Course Project

- Implemented an algorithm for motion feature extraction based on Lucas-Kanade optical flow to identify vehicle contour lights.
- Analyzed the impact of optical flow thresholds on noise filtering and target recognition stability for intelligent traffic detection.

Skills

Expertise: Distributed Optimization & Primal-Dual Algorithms, Multi-agent System Modeling, Numerical Solution and Simulation of PDEs, Control System Design (PID/PI)

Languages: Mandarin (Native), English (CET-6), Japanese (JLPT N2)

Programming: Python (Pandas, PyTorch, Sklearn, OpenCV), MATLAB, Java, LaTeX/Markdown

Honors and Awards

First-Prize Graduate Academic Scholarship, Southwest University	Jan 2025
Tianji Annual Scholarship, Donghua University	Jan 2024
Honorable Mention, MCM/ICM	Jan 2023
Third-Prize Undergraduate Academic Scholarship, Donghua University	2022 & 2023
Third Prize (Shanghai Division), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	Jan 2023
Third Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM)	Jan 2023